UNIT II : Data link layer

In the TCP/IP model, the Data Link Layer (also called the Link Layer or Network Access Layer) is responsible for preparing IP packets for transmission over the physical network by encapsulating them into frames. It provides reliable communication between two directly connected devices on the same network. It uses the services of the Physical Layer to transmit and receive bits over the communication medium.

Design issues of Data link Layer

It has a number of functions, including:

1. Providing a well-defined service interface to the network layer.
2. Dealing with transmission errors.
3. Regulating the flow of data so that slow receivers are not swamped by fast senders.

To accomplish these goals, the data link layer takes the packets it gets from the network layer and encapsulates them into frames for transmission. Each frame contains a frame header, a payload field for holding the packet, and a frame trailer, as illustrated in Fig.

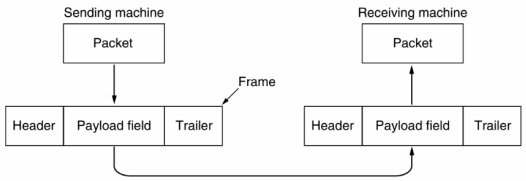


Figure 3-1. Relationship between packets and frames.

Framing:

Framing is the process of dividing the data received from the Network Layer into smaller units called frames by adding a header and trailer. Each frame is then transmitted over the physical communication medium.

Four Types of framing methods: are

1. Byte count.
2. Flag bytes with byte stuffing.
3. Flag bits with bit stuffing.
4. Physical layer coding violations.

Byte Count Framing

In the **Byte Count** method, the frame header contains a field that specifies the number of bytes in the frame. The receiver uses this count to determine where the frame ends. However, if the byte count is corrupted due to a transmission error, the receiver loses synchronization and cannot identify the start of the next frame. Therefore, this method is rarely used alone.

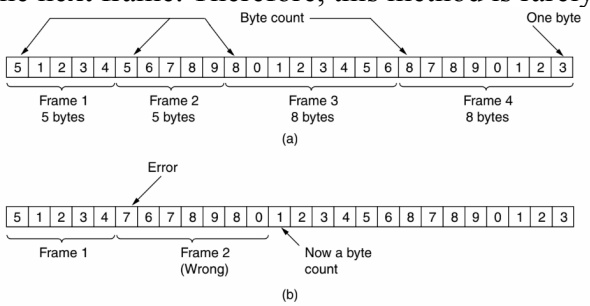


Figure 3-3. A byte stream. (a) Without errors. (b) With one error.

2. Flag bytes with byte stuffing (Byte-Oriented Framing with Flag Bytes) :

In this method, each frame starts and ends with a special flag byte (FLAG)**.** Two consecutive flag bytes indicate the end of one frame and the beginning of the next**.** If synchronization is lost, the receiver can detect the next frame by searching for the flag bytes, making frame recovery easier.

Byte Stuffing : If a flag byte (FLAG**)** appears in the data, it may be mistaken for a frame boundary. To avoid this, the sender inserts an escape byte (ESC) before every flag byte in the data. The receiver removes the escape bytes before delivering the data to the network layer. This technique is called Byte Stuffing. Byte stuffing is the process of adding 1 extra byte whenever there is a flag or escape character in the text.

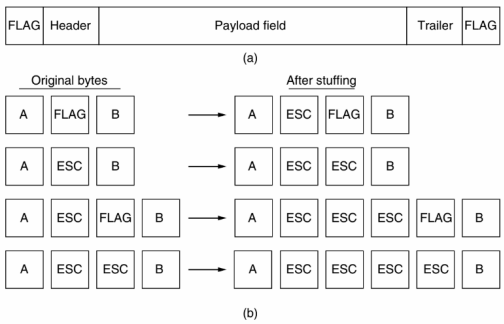
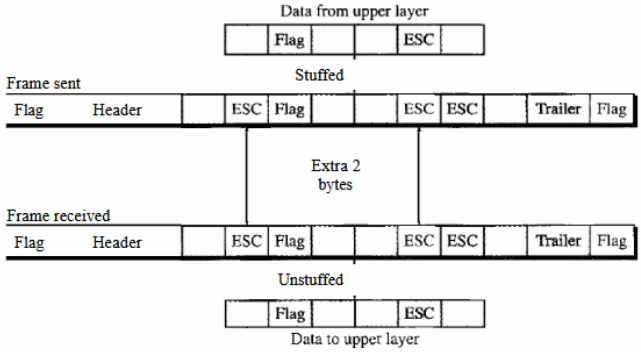
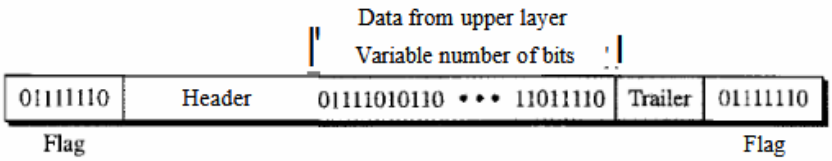


Figure 3-4. (a) A frame delimited by flag bytes. (b) Four examples of byte sequences before and after byte stuffing.

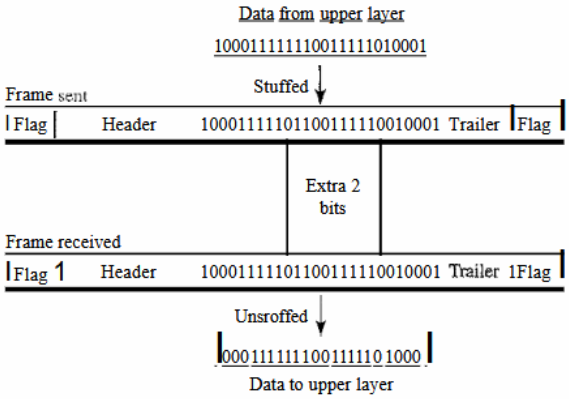


3. Flag bits with bit stuffing.

This framing technique that works at the bit level instead of the byte level. This method was developed for the HDLC (High-Level Data Link Control) protocol. Each frame begins and ends with the special bit pattern 01111110 **(**0x7E), called flag**.** the



Bit Stuffing : Whenever the sender detects five consecutive 1s in the data, it inserts a 0 bit automatically. This prevents the flag pattern from appearing within the data and ensures correct frame identification. This technique is called Bit Stuffing. When the receiver sees five consecutive incoming 1 bits, followed by a 0 it,it automatically destuffs (i.e., deletes) the 0 bit.



MCQ:

1. Which layer is responsible for framing?

A) Physical Layer B) Network Layer C) **Data Link Layer** D) Transport Layer

2. Which framing technique uses the first field to indicate the frame length?

A) Bit Stuffing B) Character Stuffing C) **Byte Count** D) Physical Layer Coding Violation

3. In Byte Stuffing, which character is inserted before a special flag character?

A) ACK B) NAK C) **ESC (Escape)** D) SYN

4. In Bit Stuffing, a '0' bit is inserted after

A) Three consecutive 1s , B) Four consecutive 1s, C) **Five consecutive 1s** , D) Six consecutive 1s

5. Which flag pattern is commonly used in HDLC for frame delimiting?

A) 11111111 B) 00000000 C) 01111110 D) 10101010

1. The process of dividing a bit stream into identifiable units is called _____. **Answer:** Framing

2. In the Character Count method, the first field of the frame indicates the _____ of the frame. **Answer:** Number of characters (or Frame length)

3. In Character Stuffing, the _____ character is inserted before a special flag character. **Answer:** ESC (Escape)

4. In Bit Stuffing, a _____ bit is inserted after every five consecutive 1s.**Answer:** 0

5. The standard flag pattern used in Bit Stuffing is _____. **Answer:** 01111110